

Risk-Aware Water-Surface Obstacle Detection and Warning via Segmentation and Monocular Depth Estimation

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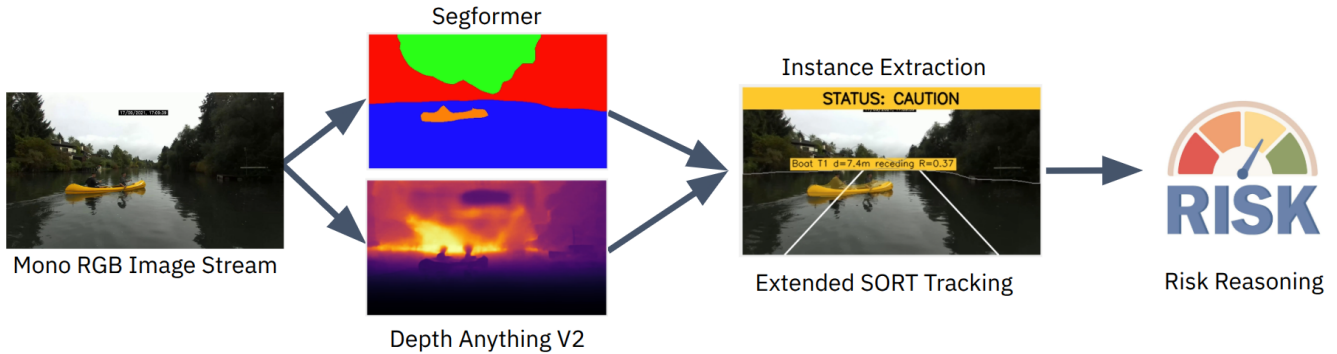


Fig. 1: Pipeline of the system.

Abstract—Safe navigation of Autonomous Surface Vehicles (ASVs) requires not only accurate obstacle detection but also actionable risk assessment. We present a vision-only, risk-aware framework that fuses semantic segmentation (SegFormer) with radar-calibrated monocular depth estimation (Depth Anything V2) using a single RGB stream. To translate raw perception into practical safety alerts, we introduce a continuous collision-corridor risk model that evaluates forward proximity, lateral offset, closing velocity, and semantic class hazard. An extended SORT tracker jointly estimates bounding-box dynamics and metric depth, ensuring temporally stable Time-to-Collision predictions. Evaluated on real-world maritime sequences, the system operates at approximately 13.7 FPS, successfully suppressing frame-to-frame noise to provide reliable, discrete safety warnings (SAFE, CAUTION, IMMEDIATE) for low-cost ASV platforms.

I. INTRODUCTION

Autonomous Surface Vehicles (ASVs) are increasingly being deployed in complex and unstructured maritime environments, such as near-shore ports, marinas, and inland waterways. These domains present significant perception challenges, characterized by highly dynamic water surfaces, strong reflections, waves, and the frequent presence of low-profile or partially submerged obstacles such as buoys, swimmers, and floating debris. Unlike structured urban roadways with consistent ground planes, the maritime environment lacks static reference features, which complicates standard computer vision tasks. Failure to detect and respond to these hazards in

a timely manner can lead to severe safety incidents, including collisions and groundings.

To enable safe navigation, conventional ASV platforms heavily rely on specialized sensing modalities such as marine radar, LiDAR, or sonar. While effective, these sensors significantly increase system cost, power consumption, and hardware complexity, making them impractical for small-scale or low-cost autonomous platforms. Monocular cameras offer a lightweight and inexpensive alternative; however, current vision-based maritime perception research has primarily focused on improving pixel-level accuracy metrics, such as mean Intersection-over-Union (mIoU).

In safety-critical navigation, a critical gap exists between perception accuracy and operational safety: high segmentation accuracy does not inherently translate into safe navigation. The fundamental question for an ASV is not merely whether an obstacle is perfectly segmented, but a more urgent one: “Does this obstacle pose an immediate collision risk?” Standard vision systems often provide raw detections that lack this risk-aware context, potentially overwhelming control modules with data while failing to highlight the most pressing threats.

Motivated by these challenges, this project proposes a risk-aware, camera-only obstacle detection and warning framework via semantic segmentation and monocular depth estimation. Rather than producing raw detections, our approach aims to generate human-interpretable and actionable warnings. By

leveraging SegFormer for robust maritime semantic segmentation and Depth Anything V2 for proximity estimation, we fuse spatial semantics with depth cues inferred from a single RGB stream. To ensure physical groundedness, we calibrate the monocular depth estimates to a metric scale using radar ground truth.

By reasoning about obstacle proximity, lateral threat, and semantic class, our system transitions from raw perception to risk assessment, generating discrete warning signals: SAFE, CAUTION, and IMMEDIATE WARNING. The framework is systematically evaluated on real-world maritime sequences, prioritizing safety-oriented metrics over traditional segmentation accuracy alone, ensuring that the system is both reliable and viable for real-time deployment on resource-constrained ASVs.

II. RELATED WORK

Our system integrates four lines of prior work: maritime semantic segmentation, monocular depth estimation, collision-risk modelling, and multi-object tracking. We summarise the most relevant contributions below and clarify how our design differs.

A. Maritime Semantic Segmentation

General-purpose semantic segmentation has been driven by fully-convolutional architectures such as U-Net [1] and the DeepLab family [2], which introduced atrous convolutions and spatial-pyramid pooling to enlarge the effective receptive field. More recently, transformer-based models such as SegFormer [3] have shown that hierarchical self-attention can match or surpass CNN baselines while remaining lightweight and robust to appearance variation.

In the maritime domain, WaSR [4] and MODD [5] introduced a dedicated water segmentation and refinement network tailored to cope with wakes, reflections, and sun glitter. Training data for ASV perception has been consolidated in dedicated datasets such as LaRS [6], a diverse panoptic benchmark spanning many vessels, weather conditions, and obstacle categories, which we use as our primary training and evaluation source. Complementary multi-modal datasets such as WaterScenes [7] pair camera imagery with 4D radar to support sensor-fusion research.

Most of this literature pursues higher pixel-level accuracy (mIoU) as the primary objective. Our work instead treats segmentation as one component of a downstream warning pipeline and therefore re-scales evaluation around safety-oriented metrics (obstacle recall and inference latency). We also extend SegFormer with a boundary-prediction head, which is not present in the original paper, to recover instance separation from a semantic-only backbone.

B. Monocular Depth Estimation

Monocular depth estimation has shifted from dataset-specific regressors to generalist foundation models. MiDaS [8] demonstrated that mixing multiple depth datasets enables zero-shot relative-depth prediction across domains. Depth Anything [9] further scales this idea by training on tens of millions

of unlabeled images with a semi-supervised teacher–student pipeline, while Depth Anything V2 [10] improves boundary sharpness by combining synthetic training data with a larger teacher. Diffusion-based alternatives such as Marigold [11] and DepthFM [12] have recently shown strong fine-detail recovery at the cost of slower inference.

All of these models are trained primarily on road-driving and indoor imagery [13]. When deployed on open-water scenes, they exhibit a consistent domain gap: the horizon saturates, and metric scale drifts away from physical units. Rather than retraining a large depth model on scarce maritime data, we adopt a lightweight calibration strategy: the off-the-shelf metric-outdoor variant of Depth Anything V2 is paired with a single global scaling factor obtained from radar ground truth. This keeps the pre-trained depth representation intact while restoring near-range metric accuracy, which is what matters most for close-in collision warning.

C. Collision Risk and Warning Systems

Early warnings in autonomous driving have followed two lines of thought. The first is *formal safety*, exemplified by Mobileye’s Responsibility-Sensitive Safety (RSS) framework [14], which derives longitudinal and lateral safe distances from response time and braking capability, and by the ISO 22839 standard for forward collision mitigation [15]. The second line is *domain-specific heuristics*. In the marine domain, the ship-domain concept of Fujii and Tanaka [16] sizes a safe region around a vessel from its beam and operating speed, and more recent ASV work adapts COLREGs rules to reactive collision avoidance [17].

Our risk model borrows the *geometric decomposition* of RSS, separating proximity into a forward-range term and a lateral-excess term relative to a corridor, but replaces the kinematic safe-distance equations with a continuous exponential-decay score. Unlike binary safe/unsafe formulations, our score is continuous, weighted by obstacle class, and further modulated by a closing-velocity amplification factor, which allows the system to distinguish between approaching and receding obstacles at the same range.

D. Multi-Object Tracking

SORT [18] is a widely-used online tracker that combines a constant-velocity Kalman filter on 2D bounding boxes with Hungarian matching on IoU. DeepSORT [19] augments SORT with a learned appearance embedding to reduce identity switches through occlusion. We adopt the SORT formulation for its low latency and minimal training requirements, but extend its 7-dimensional Kalman state to nine dimensions so that metric depth and its rate of change are tracked jointly with bounding-box dynamics.

E. Summary of Contributions

Against this background, the main contributions of our design are: (i) an end-to-end vision-only perception pipeline in which segmentation and monocular depth are fused through connected-component instance extraction, avoiding specialised

ranging hardware; (ii) a radar-calibrated global scaling strategy that makes a terrestrial pre-trained depth model metrically usable on water within the close-in range that matters for collision warning; (iii) a continuous collision-corridor risk score that unifies RSS-style geometric reasoning with ship-domain sizing, semantic class weighting, and closing-velocity amplification; and (iv) a depth-augmented SORT tracker that jointly estimates bounding-box motion and closing velocity, enabling stable Time-to-Collision estimates despite noisy per-frame depth.

III. TECHNICAL APPROACH

Our system follows a modular perception pipeline consisting of three stages: *Semantic Segmentation*, *Monocular Depth Estimation*, and *Risk-Aware Sensor Fusion with Tracking*. The first two stages run in parallel on each RGB frame; their outputs are fused into per-obstacle risk scores, smoothed over time by an extended SORT tracker, and mapped to discrete warning levels. Figure 1 illustrates this flow on a representative image in the MODD dataset.

A. Semantic Segmentation

1) *Dataset*: We train and evaluate our segmentation model on the LaRS dataset [6], a diverse panoptic maritime benchmark containing real-world surface-navigation imagery under challenging visual conditions such as water reflections, sun glitter, and partially submerged obstacles. LaRS provides both semantic and instance-level annotations, which we leverage to train the dual-head model described below.

2) *Instance-Aware Dual-Head Model*: Our segmentation backbone is SegFormer [3] with the MiT-B0 encoder, chosen for its favourable accuracy–latency trade-off among the architectures we benchmarked. To enable instance-level obstacle separation for the downstream risk module, we extend the original single-head architecture with a *dual-head design*: a semantic head producing 9-class logits and an auxiliary boundary head predicting a per-pixel obstacle-boundary probability map. The 9 classes, Static Obstacle, Water, Sky, Boat, Buoy, Swimmer, Animal, Float, and Other, are obtained by merging the LaRS panoptic categories. The model is trained with a combined loss of cross-entropy, Dice, and binary cross-entropy on boundary targets, with class-specific weights to counter severe class imbalance (e.g., Swimmer 5 $\times$, Buoy 4 $\times$). At inference, the boundary map is used to split adjacent obstacle instances of the same class during connected-component extraction (Section III-C1).

B. Monocular Depth Estimation

1) *Dataset*: We use the sample subset of the WaterScenes dataset [7] for depth calibration and evaluation. WaterScenes provides synchronised camera and 4D-radar streams. We use only the RGB images as input and the radar range as metric ground truth. Radar points are projected onto image coordinates (u, v) using the provided calibration, giving per-pixel distance samples to compare against monocular predictions.

2) *Calibration to the Maritime Domain*: We adopt the Metric-Outdoor variant of Depth Anything V2 [10], which is fine-tuned from the foundation model on synthetic Virtual KITTI data [13]. Because Virtual KITTI is dominated by road-driving scenes, directly applying the pre-trained model to water-surface imagery leads to a consistent scale bias. Rather than retraining on scarce maritime depth data, we correct this with a single global scaling factor obtained by fitting against WaterScenes radar ground truth:

$$d_{\text{cal}} = s \cdot d_{\text{raw}}, \quad s = 0.4. \quad (1)$$

This preserves the foundation model’s representation while restoring metric scale in the close-in range that matters most for collision warning.

3) *Limitations and Mitigation*: Even after calibration, monocular depth accuracy degrades noticeably beyond 20 m, and frame-to-frame estimates exhibit residual noise (median inter-frame variation of 0.28 m on our evaluation sequences). Rather than pursuing further calibration, the downstream modules are designed to tolerate this uncertainty: the risk model uses conservative depth percentiles (p5) per instance, the Kalman tracker smooths depth over time, and warning-level hysteresis suppresses noise-induced flicker. These mechanisms are detailed in the following sections.

C. Obstacle Extraction and Risk-Aware Warning

1) *Instance Extraction*: The segmentation mask and calibrated depth map are fused to extract individual obstacle instances. For each non-background class (excluding Water and Sky), connected-component analysis isolates contiguous obstacle regions. When the boundary probability map is available, boundary pixels above a threshold ($\tau_b = 0.4$) are suppressed before labelling, splitting adjacent objects of the same class into separate instances. Regions smaller than 300 px are discarded as segmentation noise. For each surviving instance, a representative depth is computed as the 5th percentile of its per-pixel depth distribution, giving a conservative estimate of the nearest point of approach. The lateral position is expressed as a normalised offset $\ell \in [-1, 1]$ from the image centre.

2) *Collision-Corridor Risk Model*: Drawing on the geometric decomposition of Mobileye’s Responsibility-Sensitive Safety framework [14], which separates safety into longitudinal and lateral components, and the ship-domain sizing intuition of Fujii and Tanaka [16], we define a virtual forward collision corridor centred on the ASV heading. The corridor half-width is $W_{\text{boat}}/2 + M_{\text{lat}}$, with $W_{\text{boat}} = 2$ m (ASV beam) and $M_{\text{lat}} = 1$ m (lateral safety margin).

For each obstacle with forward distance d_{fwd} and lateral distance $d_{\text{lat}} = \ell \cdot d_{\text{fwd}} \cdot \tan(\theta_{\text{HFOV}}/2)$, the lateral excess beyond the corridor is:

$$e_{\text{lat}} = \max(0, |d_{\text{lat}}| - (W_{\text{boat}}/2 + M_{\text{lat}})). \quad (2)$$

The static risk baseline is:

$$R_{\text{base}} = w_{\text{class}} \cdot \exp(-d_{\text{fwd}}/D_{\text{safe}}) \cdot \exp(-e_{\text{lat}}/L_{\text{safe}}), \quad (3)$$

with $D_{\text{safe}} = 12$ m (forward decay, chosen within the reliable 0–20 m depth range), $L_{\text{safe}} = 2$ m (lateral decay outside the corridor), and per-class weights w_{class} reflecting inherent hazard severity (e.g., Boat 1.0, Buoy 0.7, Static Obstacle 0.3).

To incorporate relative motion, the static baseline is modulated by a velocity amplification factor derived from the Kalman-filtered closing velocity (Section III-D):

$$R = R_{\text{base}} \cdot (1 + \alpha_v \cdot \min(v_{\text{closing}}/V_{\text{ref}}, 2)), \quad (4)$$

where $V_{\text{ref}} = 3$ m/s is a reference closing speed (typical inland-vessel speed), $\alpha_v = 1.0$, and v_{closing} is clamped at zero for stationary or receding objects. The factor is capped at $2V_{\text{ref}}$ to prevent noisy velocity estimates from producing extreme amplification. When no velocity is yet available (first few frames before a track is confirmed), the factor defaults to 1.

Equation (4) has several desirable properties: (i) an obstacle directly ahead at $d_{\text{fwd}} = 0$ returns $R = w_{\text{class}}$; (ii) risk decays exponentially with distance; (iii) obstacles outside the corridor experience additional lateral decay; (iv) the multiplicative form requires both proximity *and* alignment to be threatening; and (v) approaching obstacles are penalised more heavily than stationary or receding ones.

The continuous risk score is discretised into three warning levels:

- **SAFE:** $R < 0.15$
- **CAUTION:** $0.15 \leq R < 0.50$
- **IMMEDIATE WARNING:** $R \geq 0.50$

With the velocity factor, the maximum attainable score is $w_{\text{class}} \cdot (1 + 2\alpha_v)$, so a high-weight obstacle approaching rapidly can exceed 1.0. The frame-level warning is the maximum across all obstacles.

D. Dynamic Object Tracking and Velocity Estimation

1) *SORT with Kalman-Filtered Depth:* To associate obstacle instances across frames and estimate their motion, we extend SORT [18] with a Kalman state that jointly tracks bounding-box dynamics and metric depth. Standard SORT uses a 7-dimensional state $[c_x, c_y, s, r, \dot{c}_x, \dot{c}_y, \dot{s}]$. We enlarge it to nine dimensions:

$$\mathbf{x} = [c_x, c_y, s, r, d, \dot{c}_x, \dot{c}_y, \dot{s}, \dot{d}]^\top, \quad (5)$$

where d is the calibrated metric depth and $\dot{d}$ its rate of change (m/frame). The observation vector $\mathbf{z} = [c_x, c_y, s, r, d]^\top$ includes the depth reading from the monocular depth module directly. A constant-velocity transition model predicts $d_{t+1} = d_t + \dot{d}$, and the Kalman update fuses noisy depth observations with the predicted state.

This formulation has three advantages over post-hoc velocity estimation (e.g., finite differencing or regression on a depth history buffer): (i) the filter balances observation noise and process dynamics optimally, automatically down-weighting unreliable depth measurements; (ii) the velocity $\dot{d}$ is estimated as part of the state, yielding smooth trajectories

without manual EMA or windowing; and (iii) during temporary detection loss (up to `max_age` frames), the filter predicts depth forward, maintaining velocity estimates.

Data association uses the Hungarian algorithm on an IoU cost matrix between predicted and observed bounding boxes, with a minimum IoU of 0.20. Unmatched detections spawn new tracks; tracks without matches for five consecutive frames are deleted, and a track is confirmed after two consecutive hits.

2) *Closing Velocity and Time-to-Collision:* For each confirmed track, the closing velocity is derived from the Kalman state:

$$v_{\text{closing}} = -\dot{d} \cdot f_{\text{fps}}, \quad (6)$$

where f_{fps} is the frame rate and positive values indicate approach. When $v_{\text{closing}} > 0.1$ m/s, the Time-to-Collision is:

$$\text{TTC} = d/v_{\text{closing}}, \quad (7)$$

using the Kalman-filtered depth rather than the noisy per-frame observation. The closing velocity is also fed back into the risk model (4) to amplify the risk score for approaching obstacles.

E. Temporal Smoothing

Raw frame-by-frame predictions from segmentation and depth estimation are inherently noisy, leading to fluctuating risk scores and warning-level flicker. We apply three complementary per-track smoothing mechanisms:

Lateral-offset EMA. The normalised lateral position of each tracked obstacle is smoothed with an exponential moving average ($\alpha = 0.4$), preventing small shifts in the segmentation mask from causing abrupt changes in the lateral-decay term of Eq. (4).

Kalman-filtered depth and velocity. As described above, the extended Kalman state inherently smooths both depth and closing velocity, eliminating the need for separate post-hoc filtering.

Warning-level hysteresis. To prevent rapid toggling between warning levels at the score thresholds, we use separate enter/exit thresholds: a higher level is triggered when the score exceeds the enter threshold (e.g., 0.50 for IMMEDIATE), but the level only drops back when the score falls below a lower exit threshold (e.g., 0.38). This produces stable, interpretable warning transitions in the presence of residual noise.

IV. RESULTS

A. Semantic Segmentation

1) *Quantitative Analysis:* We evaluated two semantic segmentation baselines, DeepLab and SegFormer, on the LaRS validation split. To better reflect the safety-oriented goal of the project, the primary focus was on Obstacle Recall, which serves as our proxy for “missed hazards,” and Inference Latency, which determines real-time viability.

The results in Table I indicate that SegFormer provides superior safety performance, achieving a 96.51% Obstacle Recall. This is a significant improvement over the 79.83% Recall of the DeepLab baseline, meaning SegFormer is far less likely to miss hazardous objects like buoys or debris. While

TABLE I: Comparison of DeepLab and SegFormer on the LaRS validation set.

	DeepLab	SegFormer
mIoU	0.8648	0.9659
Obstacle Recall	0.7983	0.9651
Obstacle Precision	0.9648	0.9737
Avg Latency (ms)	15.44	29.74

the SegFormer architecture exhibits a higher average latency of 29.74 ms compared to the DeepLab baseline, it remains highly viable for deployment on autonomous surface vehicles. For a boat moving at typical maritime speeds, a processing time of 29.74 ms (roughly 33 FPS) is still well within the limits for real-time safety and collision avoidance.

To further analyze class-wise segmentation performance, we report the Intersection-over-Union (IoU) for each semantic category on the evaluation set (198 images), as shown in Table II.

TABLE II: Per-class IoU on segmentation test set

Class	IoU
Static Obstacle	0.9056
Water	0.9703
Sky	0.9618
Boat	0.6230
Buoy	0.2036
Swimmer	0.4254
Animal	0.0068
Float	0.0000
Other	0.2442
mIoU	0.4823

The results show strong performance on dominant background classes such as Water (0.9703) and Sky (0.9618), as well as high accuracy on large static obstacles (0.9056). In contrast, performance drops significantly for smaller or less frequent classes such as Buoy (0.2036), Swimmer (0.4254), and especially Animal (0.0068) and Float (0.0000).

This behavior is mainly due to severe class imbalance in the training data, the small spatial extent of these objects, and their high appearance variability (e.g., partial occlusion and interaction with water), which makes accurate segmentation more challenging compared to dominant classes.

Despite the relatively low overall mIoU (0.4823), the model maintains strong obstacle recall (Table I), which is more critical for safety-oriented applications. In particular, even partially detected obstacles are sufficient for downstream risk estimation, as the system primarily relies on obstacle presence and approximate localization rather than precise pixel-level segmentation.

2) *Qualitative Visualization*: Figures 2 and 3 show a qualitative comparison of DeepLab and SegFormer on a validation image. Both models correctly segment the dominant sky and water regions, indicating that they capture the overall scene layout well. The main difference lies in the thin obstacle region near the waterline. DeepLab only recovers part of this structure and produces a more fragmented prediction, whereas SegFormer preserves a larger and more continuous obstacle region that more closely matches the ground truth.

This observation is consistent with the quantitative results, especially the stronger obstacle recall achieved by SegFormer.

B. Monocular Depth Estimation

We evaluate three Depth Anything model variants (*small*, *base*, *large*) on 33,648 radar-aligned points. The average inference time per image is 0.149 s, 0.279 s, and 0.710 s for the *small*, *base*, and *large* models, respectively. Tables III, IV, and V report performance across different distance ranges for each model.

TABLE III: Distance-binned performance for the *small* model.

Range (m)	Count	MAE (m)	Est/GT
0–10	2636	2.572	0.936
0–20	8041	4.434	0.787
0–50	19789	12.755	0.600
ALL	33648	39.210	0.441

TABLE IV: Distance-binned performance for the *base* model.

Range (m)	Count	MAE (m)	Est/GT
0–10	2636	2.298	1.118
0–20	8041	4.039	0.887
0–50	19789	11.783	0.666
ALL	33648	37.614	0.493

TABLE V: Distance-binned performance for the *large* model.

Range (m)	Count	MAE (m)	Est/GT
0–10	2636	2.365	1.260
0–20	8041	3.215	1.008
0–50	19789	9.907	0.766
ALL	33648	35.440	0.565

To directly compare the three models in the near-range regime, Table VI reports performance within 20 m.

TABLE VI: Model comparison within 0–20 m.

Model	MAE (m)	RMSE (m)	Est/GT
Small	4.434	5.719	0.787
Base	4.039	5.363	0.887
Large	3.215	4.471	1.008

The results reveal several important trends. First, all models achieve substantially better performance at short ranges, while errors increase rapidly as distance grows, indicating that long-range depth estimation remains challenging in this setting. This behavior is also reflected in Fig. 4 and Fig. 5, where predictions increasingly deviate from the ideal diagonal as distance increases. Among the three variants, the *large* model consistently achieves the best accuracy across all ranges, particularly within 20 m, where it reaches near-metric scaling ($\text{Est/GT} \approx 1$). In contrast, the *small* and *base* models tend to under-estimate depth even at moderate distances, suggesting limited capacity to recover accurate metric scale. However, this accuracy improvement comes at a significant computational

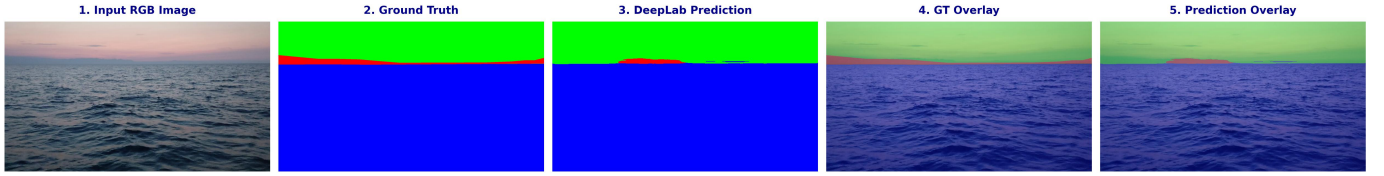


Fig. 2: Performance of DeepLab on a maritime scene.

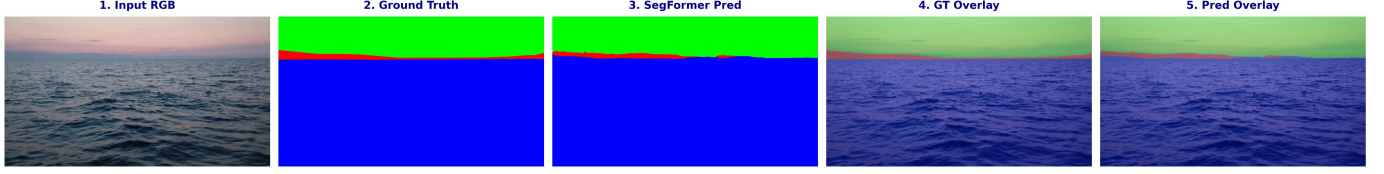


Fig. 3: Performance of SegFormer on a maritime scene.

cost, as the large model requires substantially longer inference time per image. The base model provides a reasonable trade-off between accuracy and efficiency, while the small model offers the fastest inference but the lowest accuracy. Overall, these results highlight a clear accuracy–efficiency trade-off, as well as the difficulty of generalizing monocular depth models to long-range water-surface environments.

Additionally, the Depth Anything V2 outdoor metric model imposes an upper bound on predicted depth at approximately 80m. After applying the global scaling factor (0.4), this effectively limits the maximum estimated distance to around 32m. This saturation effect is observable in both Fig. 4 and Fig. 5, where predictions flatten beyond this range. As a result, distances greater than 32m are systematically clipped, leading to large errors in long-range regions. However, if the evaluation is restricted to near-range objects (within 20m), this limitation has minimal impact, and the overall accuracy remains relatively high.



Fig. 4: Qualitative visualization of monocular depth estimation. From left to right: input RGB image with projected radar points, and the corresponding predicted depth map.

C. Pipeline Results and Performance

To evaluate the practical viability of the proposed risk-aware obstacle detection system, we integrated the semantic segmentation, monocular depth estimation, and risk modeling components into a unified end-to-end pipeline. We tested the complete framework on real-world maritime video sequences from the MODD dataset, focusing on both the qualitative accuracy of the warnings and the quantitative computational efficiency.

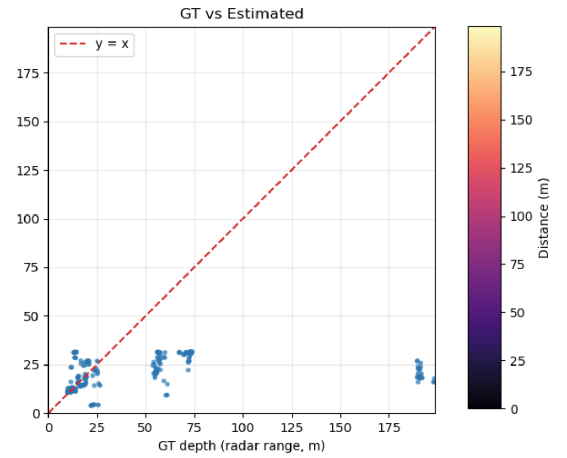


Fig. 5: Estimated depth versus radar ground truth. The deviation from the diagonal line indicates systematic underestimation, especially at longer distances.

1) *Qualitative Results on Video Sequences:* The pipeline demonstrated robust performance in dynamically assessing maritime hazards. Figure 6 presents qualitative examples of the system in operation across two different video sequences.

For each evaluated frame, the system generates a comprehensive three-pane visualization: the annotated RGB input (left), the predicted semantic mask (center), and the estimated depth map (right). The RGB frame overlays the critical outputs of our risk model, including bounding boxes, unique track IDs, distance estimates, and the computed risk score.

As shown in the examples, the system successfully distinguishes between stable and escalating risk states. In both Sequence 1 (tracking a moving boat) and Sequence 2 (tracking a static buoy), the top frames show a "SAFE" status (highlighted in green) when the objects yield a low risk score. As the relative state of the obstacles changes and the risk score increases, the system dynamically transitions into a "CAUTION" alert (highlighted in yellow, bottom frames), providing clear, actionable visual cues for potential hazards.

This confirms the effectiveness of the Mobileye RSS-based virtual collision corridor approach.

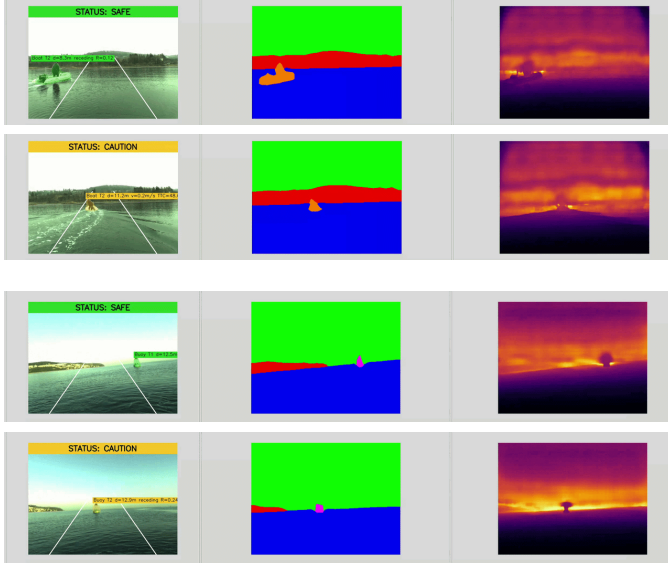


Fig. 6: Qualitative evaluation of the end-to-end pipeline on MODD video sequences. Top two rows (Sequence 1) and bottom two rows (Sequence 2) demonstrate the transition from a SAFE status to a CAUTION alert. Each visualization includes the annotated RGB frame (left), semantic segmentation (center), and monocular depth map (right).

2) *Quantitative Performance Analysis:* Beyond accurate risk representation, a critical requirement for autonomous surface vehicle (ASV) perception is the ability to process incoming data with minimal delay. To assess the real-time viability of the system, we measured the inference latency of the complete pipeline.

TABLE VII: Average inference latency breakdown of the end-to-end pipeline.

Pipeline Component	Avg Latency (ms)
Semantic Segmentation (SegFormer)	26.9
Depth Estimation (Depth Anything V2)	38.7
Temporal Fusion & Risk Modeling	7.4
Total System Latency	73.0

As detailed in Table VII, the total system latency is approximately 73.0 ms per frame, yielding an operating frequency of roughly 13.7 FPS on an Nvidia 4060 GPU. While Depth Anything V2 accounts for the majority of the computational load (53%), the extremely lightweight nature of the fusion and risk calculation step (10%) keeps the overall speed sufficient for real-time safety navigation at typical ASV speeds.

Furthermore, we evaluated the warning level distribution across 419 tested frames containing tracked obstacles. The system categorized 62% of instances as SAFE, 33% as CAUTION, and 5% as IMMEDIATE WARNING. The integration of Kalman-filtered depth and Extended SORT tracking significantly improved temporal stability, ensuring that the

safety warnings remained consistent across consecutive frames without erratic flickering.

V. CONCLUSION

In this work, we presented a lightweight, vision-only perception framework designed to bridge the gap between raw pixel-level classification and actionable safety assessment for Autonomous Surface Vehicles (ASVs). By fusing semantic masks from a dual-head SegFormer with radar-calibrated depth maps from Depth Anything V2, our system demonstrates that near-range metric accuracy can be achieved without the cost and complexity of high-end ranging hardware. The integration of an extended SORT tracker and an RSS-inspired risk model effectively smooths temporal noise, providing stable and intuitive safety warnings at a real-time rate sufficient for maritime navigation.

Moving forward, several avenues remain to be explored to enhance the system's robustness and operational scope. To overcome the scarcity of annotated maritime hazard data, we intend to construct a high-fidelity synthetic evaluation set by using generative AI video models to create diverse maritime scenarios and attaching frame-aligned, ground-truth risk labels. This will be complemented by field deployment onto an actual ASV platform to validate the system's performance against physical perturbations such as vessel heave and pitch under diverse environmental conditions. Furthermore, extending the architecture to a multi-camera configuration would provide 360-degree situational awareness through cross-camera tracking and a unified risk map. Finally, we aim to integrate these continuous risk scores directly into the ASV's path-planning and control modules, transitioning from a purely passive warning system to an active, risk-aware control loop capable of autonomous evasive maneuvers proportional to the detected threat level. Overall, this project provides a viable foundation for low-cost, intelligent maritime perception, moving toward a more comprehensive and automated understanding of operational safety.

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